

A Wronskian Identity for Antisymmetrized Sine Sums: the surface-positivity problem as sine-series ratio monotonicity, and a counterexample to the naive monotonicity principle

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Abstract

For real arrays $(a_m)_{m \geq 1}$, $(b_m)_{m \geq 1}$ (finitely supported, or summable with $\sum m(|a_m| + |b_m|) < \infty$) we record the exact identity

$$\sum_{1 \leq m < n} (a_m b_n - a_n b_m) [(m-n) \sin((m+n)t) + (m+n) \sin((n-m)t)] = 2 [F'_a(t) F_b(t) - F_a(t) F'_b(t)],$$

where $F_x(t) = \sum_m x_m \sin(mt)$: the antisymmetrized double sum is a Wronskian of two sine series. The proof is two lines. As a consequence, the open surface-positivity problem of the companion φ -lemma note — the sign of exactly such a double sum built from modified-Bessel coefficient arrays — is *equivalent* to the strict monotonicity of a ratio of two explicit sine series, and high-precision numerics indicate that the true statement is global: the Wronskian is negative on all of $(0, \pi)$, not merely near $t = \pi$. One might hope this follows from the companion result that the coefficient ratio a_m/b_m is strictly increasing, in analogy with the Biernacki–Krzyż principle for power series. It does not: we give a hand-checkable counterexample — $a = (3, 4, 3)$, $b = (3, 2, 1)$, coefficient ratio $(1, 2, 3)$ strictly increasing, b strictly decreasing so that $F_b > 0$ on $(0, \pi)$ by the classical Fejér–Gronwall positivity theorem, yet the Wronskian at $t = 2\pi/3$ equals $3\sqrt{3}/2 > 0$ — so no analogue of Biernacki–Krzyż holds for sine series, and the surface problem cannot be closed by coefficient-ratio monotonicity alone. We state the resulting conjecture precisely, record the numerical phenomenology (near-perfect parity-mirror cancellation of relative size $e^{-c\beta}$, $c \approx 2.1$, with a working-precision warning), and identify the structural route we consider most promising. The identity is machine-checked in Lean 4/Mathlib (finite form, arbitrary K ; no `sorry`, standard axiom oracle), and the counterexamples are exact algebraic evaluations; nothing beyond the identity and the counterexamples is claimed.

1 The identity

Theorem 1 (Wronskian identity). *Let (a_m) , (b_m) be real arrays, finitely supported (or with $\sum_m m(|a_m| + |b_m|) < \infty$, in which case all series below converge absolutely and uniformly). Set $F_a(t) = \sum_{m \geq 1} a_m \sin(mt)$ and likewise F_b . Then, for all $t \in \mathbb{R}$,*

$$\sum_{1 \leq m < n} (a_m b_n - a_n b_m) [(m-n) \sin((m+n)t) + (m+n) \sin((n-m)t)] = 2 [F'_a(t) F_b(t) - F_a(t) F'_b(t)]. \quad (1)$$

Proof. The product-to-sum formulas give, for any m, n ,

$$(m-n) \sin((m+n)t) + (m+n) \sin((n-m)t) = 2 [m \cos(mt) \sin(nt) - n \sin(mt) \cos(nt)] =: 2G(m, n),$$

and G is antisymmetric with $G(n, n) = 0$. Since $(a_m b_n - a_n b_m)$ is antisymmetric as well,

$$\sum_{m < n} (a_m b_n - a_n b_m) 2G(m, n) = 2 \sum_{m, n} a_m b_n G(m, n) = 2 \left[\sum_m a_m m \cos(mt) \sum_n b_n \sin(nt) - \sum_m a_m \sin(mt) \sum_n b_n n \cos(nt) \right]$$

which is $2[F'_a F_b - F_a F'_b]$. $\square$

Remark 2. Equivalently: wherever $F_b \neq 0$, the left side of (1) equals $2F_b^2 \cdot (F_a/F_b)'$. Sign questions about the antisymmetrized double sum are exactly monotonicity questions about the ratio of the two sine series — with the Wronskian form remaining meaningful (and division-free) across zeros of F_b .

2 The surface-positivity problem, reduced

We recall the objects of the companion note [4] (which proved the coefficient-ratio monotonicity used below; the wider programme context is in [6]). For $\beta > 0$ let $r_k = I_{k+1}(\beta)/I_1(\beta)$ with I_μ the modified Bessel function of the first kind, and for $k \geq 1$

$$A_k = r_{k-1}^2 \sum_{j \in \{k-2, k\}, j \geq 0} (j+1) r_j^2, \quad B_k = k r_{k-1}^4, \quad c_{mn} = A_m B_n - A_n B_m.$$

The arrays satisfy the summability hypothesis of Theorem 1 for every $\beta > 0$: by the standard large-order asymptotics $I_k(\beta) \sim (\beta/2)^k/k!$ [3, 10.41.1], the ratios r_k decay super-exponentially in k , hence so do A_m and B_m , and $\sum_m m(A_m + B_m) < \infty$. The open positivity problem is $W(t) < 0$ for t near π , where W is the left side of (1) with $(a, b) = (A, B)$. By Theorem 1,

$$W(t) = 2 [F'_A(t) F_B(t) - F_A(t) F'_B(t)], \quad F_A(t) = \sum_{m \geq 1} A_m \sin(mt), \quad F_B(t) = \sum_{m \geq 1} B_m \sin(mt).$$

Two facts sharpen the picture. First, the companion note proved $c_{mn} < 0$ for all $m < n$ (the coefficient ratio $\varphi_m = A_m/B_m$ is strictly increasing) [4]. Second, high-precision numerics (mpmath, 30–100 significant digits, truncation-stable) indicate that the Wronskian is strictly negative on *all* of $(0, \pi)$, at every β tested ($\beta \in \{0.5, 2, 8, 32\}$, grids of t/π from 0.05 to 0.99), and that $F_B > 0$ on $(0, \pi)$. The ε -window form of the original problem is thus the boundary shadow of a global statement, which we now record as the target.

Conjecture 3 (global ratio monotonicity). For every $\beta > 0$: (i) $F_B > 0$ on $(0, \pi)$, and (ii) F_A/F_B is strictly decreasing on $(0, \pi)$. Given (i), part (ii) is equivalent to $W(t) < 0$ on $(0, \pi)$, by $W = 2F_B^2 (F_A/F_B)'$; the Wronskian form $W < 0$ is the division-free statement and is what the numerics test.

3 The naive principle fails

For power series with positive coefficients, the Biernacki–Krzyż principle [1] states that if a_m/b_m is increasing then $\sum a_m x^m / \sum b_m x^m$ is increasing on $(0, 1)$. Given $\varphi_m = A_m/B_m$ increasing and $W < 0$ numerically, it is tempting to conjecture the sine-series analogue: *coefficient ratio increasing $\Rightarrow$ ratio of sine series decreasing on $(0, \pi)$* (the sign flip suggested by the boundary behaviour at $t = \pi$). This is false, even under strong positivity side conditions.

Proposition 4 (counterexample, hand-checkable). *Let $a = (3, 4, 3)$ and $b = (3, 2, 1)$ (all further entries zero). Then: the coefficient ratio $a_m/b_m = (1, 2, 3)$ is strictly increasing; b is strictly decreasing, so $F_b(t) = 3 \sin t + 2 \sin 2t + \sin 3t > 0$ on $(0, \pi)$ by the Fejér–Gronwall positivity theorem for sine polynomials with decreasing coefficients [2]; and yet at $t = 2\pi/3$,*

$$F_a = -\frac{\sqrt{3}}{2}, \quad F_b = \frac{\sqrt{3}}{2}, \quad F'_a = \frac{7}{2}, \quad F'_b = -\frac{1}{2}, \quad F'_a F_b - F_a F'_b = \frac{7\sqrt{3}}{4} - \frac{\sqrt{3}}{4} = \frac{3\sqrt{3}}{2} > 0.$$

Hence F_a/F_b is strictly increasing at $t = 2\pi/3$: no Biernacki–Krzyż analogue holds for sine series.

Proof. At $t = 2\pi/3$: $\sin t = \frac{\sqrt{3}}{2}$, $\sin 2t = -\frac{\sqrt{3}}{2}$, $\sin 3t = 0$; $\cos t = -\frac{1}{2}$, $\cos 2t = -\frac{1}{2}$, $\cos 3t = 1$. Then $F_a = (3 - 4)\frac{\sqrt{3}}{2} = -\frac{\sqrt{3}}{2}$, $F_b = (3 - 2)\frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$, $F'_a = 3(-\frac{1}{2}) + 8(-\frac{1}{2}) + 9 = \frac{7}{2}$, $F'_b = 3(-\frac{1}{2}) + 4(-\frac{1}{2}) + 3 = -\frac{1}{2}$; combine. $\square$

An even smaller example, with the Wronskian form: $a = (1, 2, 3, 4)$, $b = (1, 1, 1, 1)$ gives $F'_a F_b - F_a F'_b = 4$ at $t = \pi/2$ (there F_b vanishes, so this violates $W \leq 0$ though not ratio-monotonicity on a domain of positivity — which is why Proposition 4 carries the decreasing- b condition).

Remark 5 (the moral, in the spirit of the parity barrier). As with the parity barrier for decoupling inequalities [5], a bounded amount of structural data — here, the monotonicity of the coefficient ratio, even together with positivity of F_b — cannot certify the global sign. Whatever proves Conjecture 3 must consume more of the Bessel structure of (A_m, B_m) than the order of φ_m . This is consistent with the quantitative phenomenology: the even- and odd-parity halves of $W(\pi - \varepsilon)$ cancel to relative accuracy $\approx e^{-c\beta}$ with $c \approx 2.03$ – 2.17 (true totals e.g. $-2.9 \cdot 10^{-8}$ at $\beta = 8$; $-5.0 \cdot 10^{-29}$ at $\beta = 32$; $-5.1 \cdot 10^{-95}$ at $\beta = 100$), so any estimate that discards the exact structure loses the sign entirely; numerical exploration below $\approx 2.2\beta + 20$ working digits sees pure rounding noise. The route we consider most promising: r_k^2 and r_k^4 are ratios of Bessel products with integral (Schläfli/heat-kernel) representations, so F_A and F_B admit kernel representations on the circle; a positive integral representation of $-(F_A/F_B)'$ — or a total-positivity property of the array $(A; B)$ against the sine kernel — would close Conjecture 3 where coefficient-order arguments provably cannot.

4 Verification

Theorem 1 was verified symbolically (sympy) for generic coefficients at $K \leq 5$ (residual 0 after trigonometric normalisation) and at $K = 7$ by exact rational substitution (residual $< 10^{-35}$ at 40 digits, four random points), and numerically against the direct double sum for the Bessel arrays at $K = 90$, $\beta = 8$, at $t \in \{0.5, 1.5, 2.6, \pi - 0.1\}$ (ratio 1.0 to all displayed digits). The counterexample values of Proposition 4 are exact algebraic evaluations. The global sign scan of W and the parity-mirror measurements are as in the companion note. Scripts and sources are archived at

<https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME> (docs/BF2-ATTACK-NOTES.md, papers/wronskian-reduction).

Theorem 1 (finite form, arbitrary K) is moreover machine-checked in Lean 4/Mathlib (file `WronskianIdentity.lean` accompanying this note; checked with `lake env lean` against Lean 4.30.0-rc2, Mathlib commit `cd3b69b`): `bracket_eq_two_G` (the trigonometric collapse), `double_sum_eq_wronskian`

(the product structure), `pair_sum_eq_double_sum` (the antisymmetrization) and `wronskian_identity` (the theorem). The axiom oracle returns Lean’s three standard axioms for all four; no `sorry`:

```
'wronskian_identity' depends on axioms:  
  [propext, Classical.choice, Quot.sound]
```

(exit code 0; likewise for the three lemmas).

Acknowledgements

This note is part of an audit-first research programme whose records, Lean sources and numerical suites are public at <https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME>. Nothing is claimed about Conjecture 3 beyond its equivalence to the original surface-positivity problem and the failure of the naive route; nothing is claimed about four-dimensional gauge theory.

References

- [1] M. Biernacki and J. Krzyż, *On the monotony of certain functionals in the theory of analytic functions*, Ann. Univ. Mariae Curie-Skłodowska Sect. A **9** (1955), 135–147.
- [2] R. Askey and J. Steinig, *Some positive trigonometric sums*, Trans. Amer. Math. Soc. **187** (1974), 295–307.
- [3] F. W. J. Olver et al., *NIST Digital Library of Mathematical Functions*, §10.29, <https://dlmf.nist.gov/10.29>.
- [4] L. Eriksson, *A weighted Turán-type monotonicity lemma for modified Bessel functions via the calibrated Amos bound, with an application to the ordering of a surface expansion in two-dimensional lattice gauge theory*, July 2026; <https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME/tree/main/papers/phi-lemma>.
- [5] L. Eriksson, *Parity barriers for decoupling inequalities*, July 2026; <https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME/tree/main/papers/parity-barriers>.
- [6] L. Eriksson, *THE MASTER MAP: an audit-first navigation guide to the conditional construction of 4D $SU(N)$ Yang–Mills with mass gap*, ai.viXra:2602.0096, and the programme repository <https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME>.